Individual Work Log - Sherab Zangmo

Project: Workout Planner
Unit: CAB302 Agile Software Engineering

Date	22 March, 2026
Time Spent	3 hours
Tasks Completed	1. Researched existing AI workout planner applications such as MyFitnessPal and Fitbod. 2. Analysed key features including personalised workout plans, fitness tracking, and recommendation systems. 3. Developed initial project ideas for AI Workout Planner.
Contribution	1. Conducted research on existing workout planner applications to understand current features and industry standards 2. Identified key functionalities that can be adapted into the project. 3. Prepared ideas to support future team discussions.
Challenges	1. Unclear how complex AI implementation should be for the project scope. 2. Difficulty deciding between a simple rule-based system and a more advanced AI approach.
Next Steps	1. Refine project ideas and define core features. 2. Decide on level of AI integration. 3. Begin drafting detailed project proposals.
Evidence	Pages 3 - 5

Date	24 March, 2026
Time Spent	2 hours
Tasks Completed	1. Evaluated rule based and AI based approaches. 2. Created comparison of both approaches.

Contribution	Helped define a suitable technical approach.
Challenges	Balancing simplicity with AI requirements.
Next Steps	Finalise recommendation approach and add to proposal.
Evidence	Pages 5-7

Date	26 March, 2026
Time Spent	3.5 hours
Tasks Completed	1. Tested MyFitnessPal application to analyse calorie tracking, step count, and progress features. 2. Recorded observations and compared features with the proposed system. 3. Learned how to use GitHub for version control. 4. Uploaded project documents to the repository.
Contribution	1. Strengthened project design by incorporating real-world app insights. 2. Improved documentation quality through updates and refinements.
Challenges	Learning how to use GitHub effectively for uploading and organising files.
Next Steps	1. Continue updating GitHub repository with project progress. 2. Prepare for tutor feedback.
Evidence	Pages 8-9

Research on the existing workout apps and how they connect to our project scope

Project scope

1. Calorie tracker
2. Workout planner
3. Workout tracker
4. Goals

CASE STUDY: MyFitnessPal App

<u>Features</u>

1. AI Food Tracker – Track Calories and Macros
    - over 20 million foods from largest food databases
    - Macro tracker – see carbs, fat, protein, sodium, etc., breakdown
    - Water tracker to stay hydrated
2. Fitness – Track Workouts, Weight, and Progress
    - Activity Tracker – adding exercises and building own routines
    - See fitness progress (analyze the details of your diet and macros)
    - Exercise and count calories
    - Track with Wear OS
3. Workouts and meal plans
    - Customize your health and fitness goals – Weight loss, weight gain, weight maintenance, nutrition and fitness
    - Personalized dashboards
    - Add own meals/food tracker
    - Get connected – find friends and motivation in forums

    Premium:

Barcode scanning, meal scan, voice logging

Reviews Summary:

Many app users gave positive reviews regarding weight loss and fitness but user reviews indicate that the app is solely based on meals tracking rather than workout planning, i.e., calorie track helped maintain desired goals.

CASE STUDY: Fitbod Workout and Gym Planner

Features

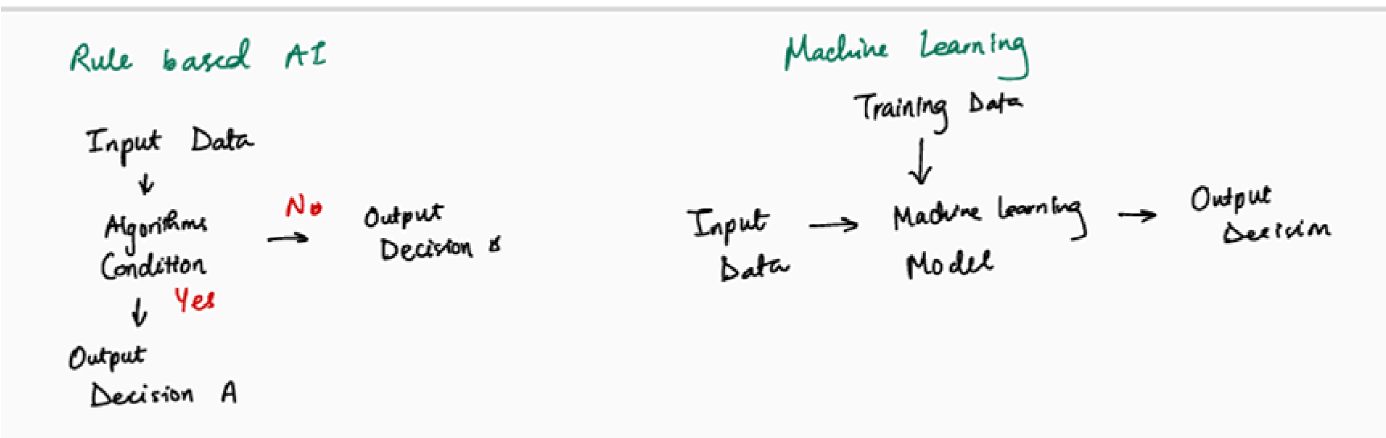
1. AI generated workouts.
2. Personalized plans based on
    - goals, fitness level and equipment
3. AI adapts workouts using:
    - Past performance
    - User feedback
4. Workout Optimizations:
    - Adjust intensity
    - Balances exercises
    - Avoids overtraining
5. Tracking Features:
    - Tracks workout history
    - Monitors progress (strength, performance)
    - Recovery tracking
6. Exercise Library
    - 1000+ exercises
    - Videos + instructions
    - Filter by muscle group and equipment
7. AI Capabilities
    - Machine learning-based recommendations.

- Adaptive learning from user behavior.
- Dynamic workout generation.

Comparison:

Fitbod focuses on AI-driven workout generation and personalized training plans, while MyFitnessPal focuses on calorie tracking and basic activity monitoring. Fitbod provides more advanced recommendation features, whereas MyFitnessPal is simpler and easier to use. For this project, a combination of both approaches can be useful, incorporating personalized workout suggestions along with basic tracking features.

<u>Rule Based Systems Model and Machine Learning Model</u>



<u>Rule Based Systems in AI</u>

These systems operate on a set of predefined rules and logic to make decisions, perform tasks or derive conclusions.

Benefits of Rule-Based Systems

1. Transparency: Rule-based systems are highly transparent because the rules governing their decisions are explicit and understandable. This clarity makes it easier to trace and debug the system's behavior.
2. Ease of Implementation: For well-defined problems with clear rules, rule-based systems are relatively easy to implement. They do not require extensive data for training, unlike machine learning models.

3. Consistency: Rule-based systems provide consistent responses and decisions as they follow predefined rules. This consistency is crucial in applications where uniformity is essential.
4. Ease of Updating: Rules can be updated or added to adapt to new knowledge or changes in the domain. This flexibility allows the system to evolve with the changing requirements.

Limitations of Rule-Based Systems

1. Scalability: As the number of rules grows, rule-based systems can become difficult to manage. The complexity of the rule base may lead to inefficiencies and increased maintenance efforts.
2. Lack of Learning Capability: Rule-based systems do not learn from new data. They rely on predefined rules and cannot adapt or improve based on experience, unlike machine learning systems that can learn and optimize over time.
3. Rigidity: Rule-based systems are inflexible when dealing with ambiguous or incomplete information. They perform best when all conditions are clearly defined, but they struggle with uncertainty and variability.
4. Difficulty Handling Complex Problems: For complex problems with interrelated factors and nuances, rule-based systems may be insufficient. They may not handle intricate patterns or relationships as effectively as advanced AI techniques.

<u>Machine Learning</u>

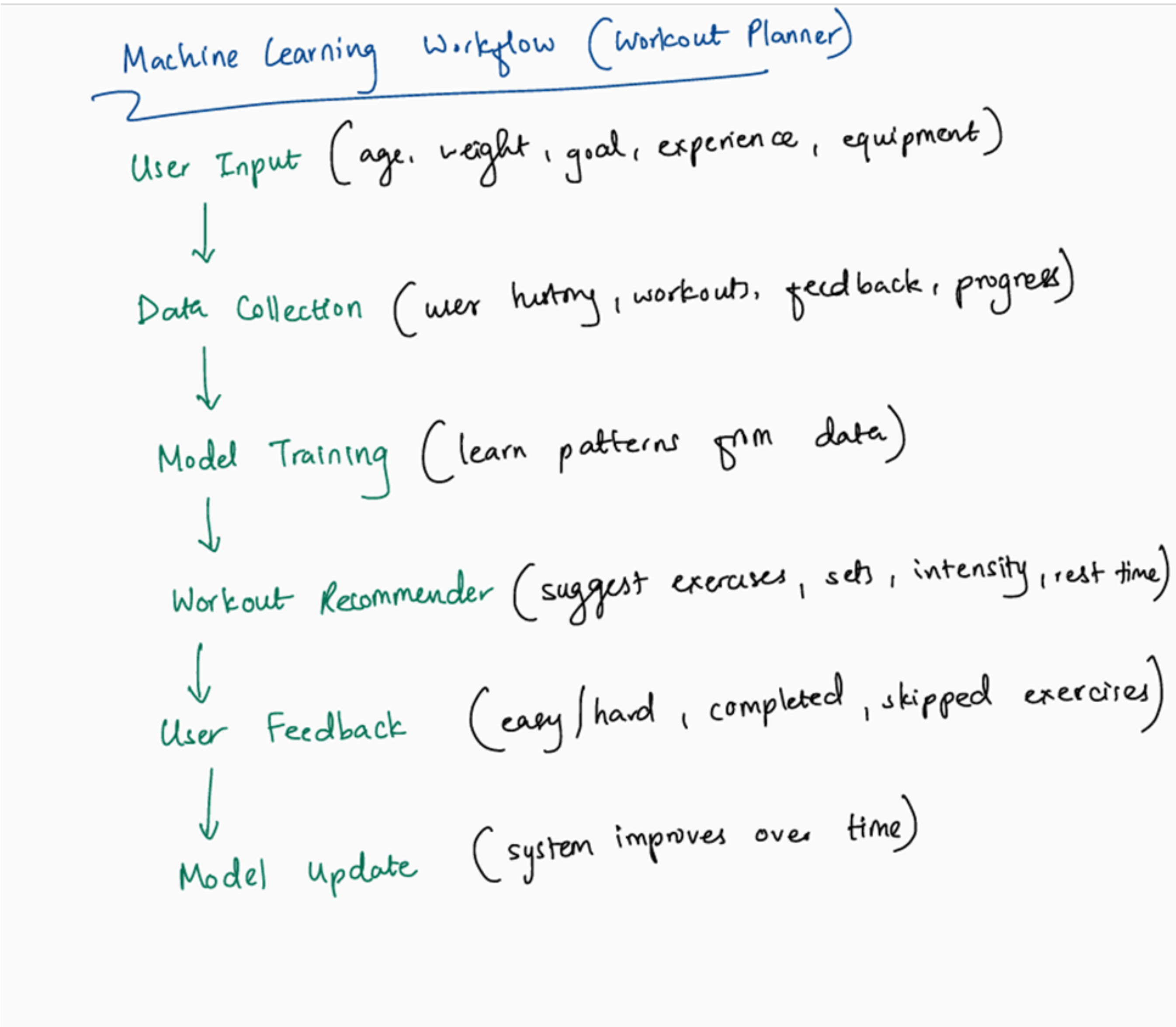
These systems use datasets (patterns and trends) to predict the target data.
(example from the application: If a user has walked more than 20k steps in a day, recommend the exercises that are not intense).

Machine learning is:

1. Less transparent (not really easy to understand)
2. Complex to develop

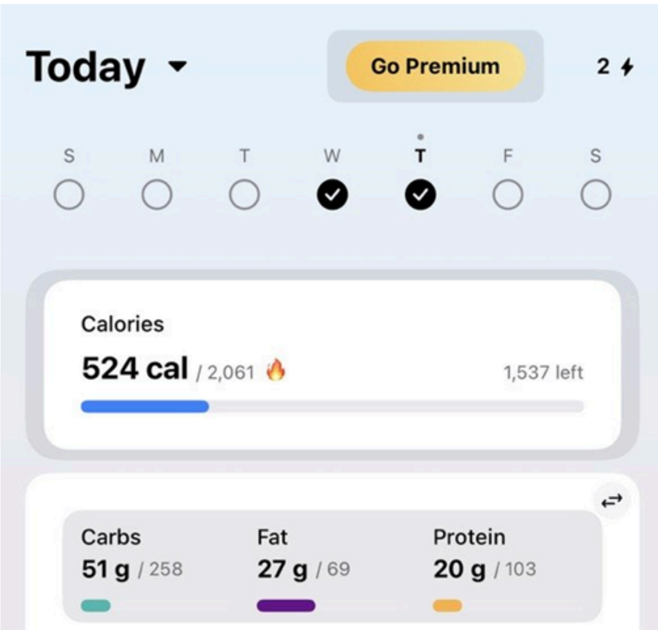
But,

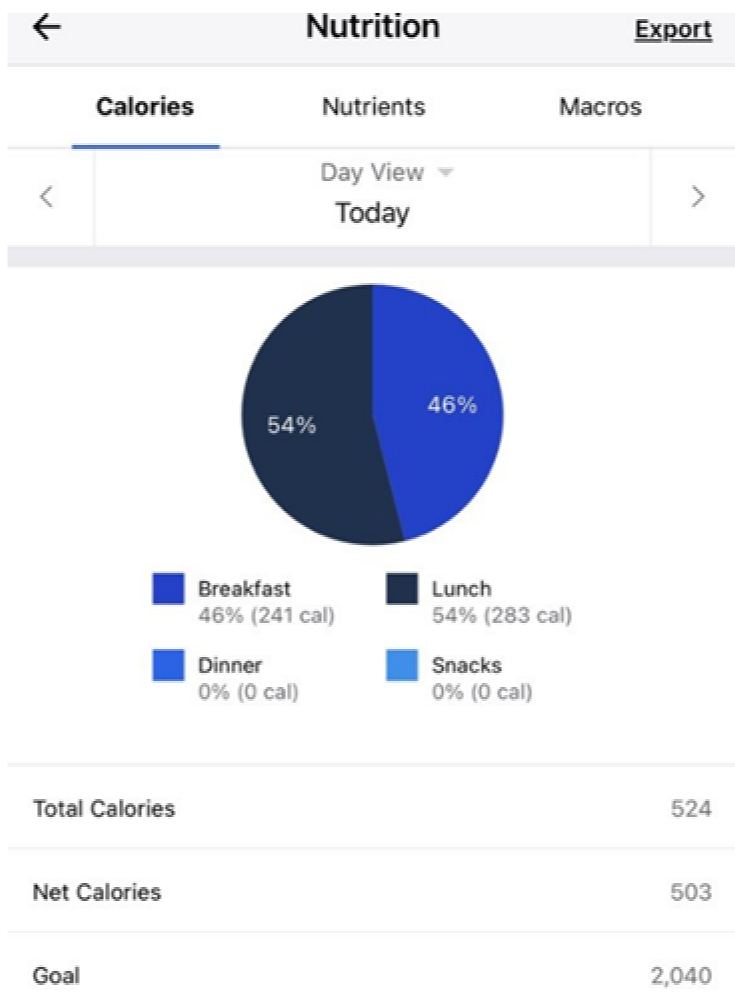
1. Its learning capability improves over time with more data.
2. Best for complex and personalised systems.



The system will use a hybrid approach combining rule based logic and machine learning. Rule based components handle basic workout structure, while machine learning adapts and personalises workouts based on user performance and feedback over time.

MyFitnessPal App Testing Evidence





Calories | **Nutrients** | Macros

Day View

Today

	Total	Goal	Left
Protein	20	103	83
Carbohydrates	51	258	207
Fiber	4	25	21
Sugar	8	77	69
Fat	27	69	42
Saturated Fat	7	23	16
Polyunsaturated Fat	3	-	-3
Monounsaturated Fat	14	-	-14
Trans Fat	0	0	0
Cholesterol			

Measurements

Steps | 1 Week

6,430	9,960	12,861
AVERAGE	BEST (25 MAR)	TOTAL

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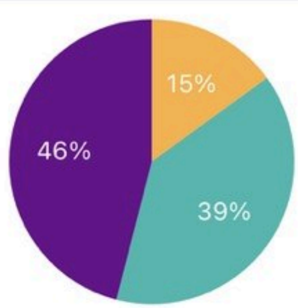
Entries

Thursday, 26 March 2026 Thursday	2,901
Wednesday, 25 March 2026 Wednesday	9,960

Calories | Nutrients

Day View

Today



	Total	Goal
Carbohydrates (51g)	39%	50%
Fat (27g)	46%	30%
Protein (20g)	15%	20%